Chiara Tramarin

MSc in Physics of Data
Vicenza, Italy

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ChiaTrama
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Education

•Master's Degree in Physics of Data Mar. 2024 – Oct. 2026
University of Padua, Padua, Italy Grade: 110/110

•Bachelor's Degree in Physics Oct. 2018 – Apr. 2024
University of Padua, Padua, Italy Grade: 92/110
Erasmus exchange: Universidad Autónoma de Madrid, Spain · Sept. 2022 – Mar. 2023

•Scientific High School Diploma 2018
Liceo P. Lioy, Vicenza, Italy

Research Experience

•Research Intern – IFIMAC, Madrid Mar. – Sept. 2026

Conducted computational research on DNA mechanics and nucleosome stability using coarse-grained molecular dynamics simulations. Performed free-energy calculations through thermodynamic integration and analyzed simulation data using Python to investigate sequence-dependent physical properties. Ran simulations on an HPC GPU cluster using Linux, Bash and SLURM.

Master's thesis: *Investigating sequence-dependent DNA mechanics and nucleosome stability through coarse-grained simulations*

•Bachelor's Thesis Research – University of Padua Nov. 2023 – Apr. 2024

Performed experimental measurements on a lithium-niobate ($LiNbO_3$) opto-microfluidic Lab-on-a-Chip for detecting transparent microplastics in water droplets, combining sample preparation, optical and electronic instrumentation (lasers, oscilloscopes and microfluidic control), and Python-based data processing and visualization.

Bachelor's thesis: *Optical sensing study of microplastics in droplets via refractive index contrast*

Co-author of a related peer-reviewed publication: *Optofluidic light-droplet interaction for rapidly assessing the presence of plastic microspheres within aqueous suspensions. Optics Express*, 34(5), 8175–8194 (2026).

Selected Academic Projects

•N-Body Simulation on CPUs and GPUs

Developed a C++ gravitational N-body simulator with serial, OpenMP/SIMD and CUDA implementations. Validated energy conservation and benchmarked CPU/GPU performance across particle counts, OpenMP thread counts and CUDA block sizes.

•Bayesian Optimization

Implemented Gaussian process-based Bayesian optimization to optimize synthetic benchmark functions and tune CNN hyperparameters (learning rate, dropout and weight decay). Compared four acquisition strategies (EI, PI, UCB and Knowledge Gradient) against random search.

•Simulation-Based Inference for a 2D Random Walk

Applied simulation-based inference with neural posterior estimation to recover parameters of a stochastic 2D random walk, validated on an analytically solvable quartic model, and extended to a red-blood-cell model not accessible analytically. Optimized simulation performance with Numba to achieve C-like speed.

•Car Image Classification with CNNs

Built and evaluated PyTorch CNNs for fine-grained vehicle classification and Siamese verification on the CompCars vehicle-image dataset, using transfer learning, fine-tuning and strategies for severe class imbalance.

•Distributed K-Means with Dask

Implemented and benchmarked a distributed K-Means‖ algorithm with Dask, extending the dask-ml implementation with weighted centroid updates. Configured VM clusters via SSH, optimized memory and thread allocation, and analyzed scalability on large datasets (KDD Cup 1999).

•Complex Networks: Social Connectedness and Self-Organized Criticality

Analyzed weighted, geolocated Facebook social-connectivity data and investigated how network structure influences collective dynamics; separately simulated self-organized criticality with the Bak–Tang–Wiesenfeld model across different network topologies.

Technical Skills & Languages

Programming Languages Python, C++, R, MATLAB, SQL
High-Performance & Parallel Computing CUDA, OpenMP, Dask, Numba, Slurm
Software & Tools Git, Docker, Linux, Jupyter, VS Code, LaTeX, Microsoft Office (Excel, Word, PowerPoint)
Languages Italian (native), English (advanced), Spanish (advanced)